

Technology News

New nanomesh may revolutionise electronics

Scientists at University of California, Los Angeles, USA, have created a graphene nanostructure called 'graphene nanomesh' that could greatly improve the functions of radios, computers, phones and other electronic devices. The new structure is able to open up a band gap in a large sheet of graphene to create a highly uniform, continuous semiconducting thin film that may be processed using standard planar semiconductor processing methods.

The nanomesh can have variable periodicities, defined as the distance between the centres of two neighbouring nanoholes. Neck widths—the shortest distance between the edges of two neighbouring holes—can be as small as 5 nanometres.

The ability to control nanomesh periodicity and neck width is very important for controlling electronic properties because charge transport properties are highly dependent on the width and the number of critical current pathways.

Using nanomesh as the semiconducting channel, the scientists have demonstrated room-temperature transistors that can support currents nearly a hundred times greater than individual graphene nanoribbon devices.

Light twists ribbons of nanoparticles

Researchers at University of Michigan, USA, have demonstrated that light can twist ribbons of nanoparticles. Calling it a rare observation, scientists said that light has been known to affect matter on the molecular scale, but bending or twisting molecules a few nanometres in size had not been seen to result in mechanical twisting of larger particles.

The discovery could be utilised in lithography or microchip production.

Nanowires developed to improve LEDs

Arizona State University researchers have developed nanowires that could make photovoltaic cells more efficient in generating power from sunlight and improve light-emitting diodes (LEDs).

The band gap of a semiconductor determines whether a given wavelength of sunlight is absorbed or left unabsorbed by the semiconductor in a solar cell. It also determines what colour of light an LED emits. To make solar cells more efficient, it's necessary to increase the range of band gaps.

The researchers improved quaternary-alloy semiconductor nanowire materials to develop new band gaps between semiconductors. This is the first time a quaternary semiconductor has been produced in the form of a nanowire or nanoparticle.

Nano device to desalinate seawater

Scientists at the Massachusetts Institute of Technology (MIT) in Cambridge,

USA, have come up with a nano device that could be used as a simple, portable water-desalination system run from a battery or solar power.

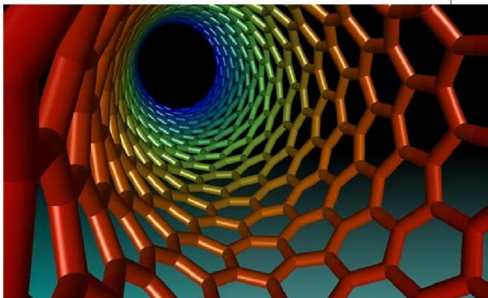
Developed by Jongyoon Han and colleagues, the device is just a few centimetres square in size, and not enough water passes through it for practical purposes.

But, according to Han, if it were possible to put many of the devices onto some kind of chip, he could produce something to rival portable household water filters.

Power from carbon nanotubes

Researchers at MIT have devised a way to produce DC voltage from carbon nanotubes. By inducing thermoelectric effect inside a carbon nanotube, electrons were found to zip through the nanotube at lightning speed, something totally unpredicted by thermoelectric physics.

By coating the nanotube's interior with different minerals, an alternating current could be created and provide a



A carbon nanotube can produce a very rapid wave of power when it is coated by a layer of fuel and ignited, so that heat travels along the tube.